English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics / SEM, CS Stream 2 / SE, Year 3
Written examination, 17 August 2023

**Problem 1.** 1. (0.25 points) A player tosses 3 fair coins and rolls 3 standard dice. For every head the player receives BGN 1, and for every six, BGN 3. What are the expected winnings?

2. (0.25 points) A player rolls a die until the sum of the numbers obtained is divisible by 6. If this happens on the $k$th roll, the player wins BGN $k$. What are the expected winnings?

3. (0.25 points) Let $X$ have a distribution on $0, 1, 2, \ldots$ such that, for $k = 1, 2, 3, \ldots$,

$$\frac{\mathbb{P}(X = k)}{\mathbb{P}(X = k-1)} = \frac{3}{k}.$$

Find the expectation and variance of $X$.

4. (0.25 points) Let the number of visitors to a stadium on a given day be $Y \sim Poi(\lambda)$. The stadium has 10 entrances $E_1, \ldots, E_{10}$, and each visitor chooses any of them with equal probability. What are the distribution, expectation and variance of the number of visitors entering through $E_1$?

**Problem 2.** $n > 2$ people each toss a fair coin. The winner is the person whose result is opposite to everyone else's. If there is no such person, the players toss again. Let $X$ be the number of rounds until a winner is determined. What are the expectation and variance of $X$?

Suppose that after the first winner is determined, the game continues until only two players remain. What is the expected number of rounds? If the $k$th winner receives $100(n-k)$, how much would you pay to participate in this game?

**Problem 3.** Let the joint density of $X$ and $Y$ be $f_{X,Y}(x, y) = cx + y$ for $x, y \geq 0$, $x + 2y \leq 1$, and 0 outside this region, where $c$ is a constant.

1. (0.25 points) Find $c$, the density of $X$, and the expectation of $Y$.

2. (0.25 points) Find $\mathbb{E}(Y \mid X = 1/2)$.

3. (0.25 points) Find the densities of the random variables $Z = X + 2Y$ and $Z = XY$.

4. (0.25 points) Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be independent and have the same distribution as $(X, Y)$. Estimate $\mathbb{P}(X_1 + \cdots + X_n > Y_1 + \cdots + Y_n)$ for large $n$.